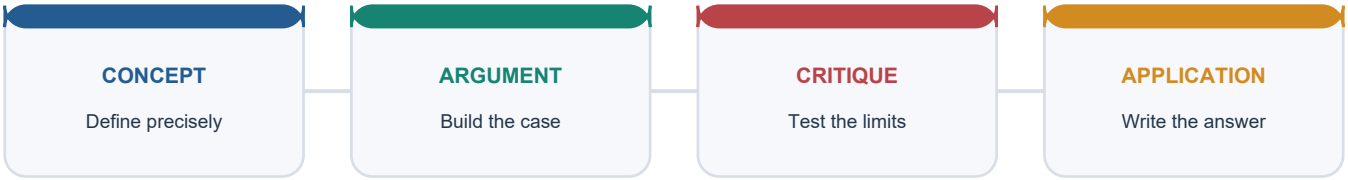
SOLVED PRACTICE WORKBOOK

Multi-Hazard Early Warning and Disaster Technology - Solved Practice Workbook

GS-III Disaster Management | 32 solved MCQs, PYQ applications and Mains models

LEARNING PATH



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Multi-Hazard Early Warning and Disaster Technology - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. A coastal district receives strong shaking before any bulletin. What is the correct warning principle?

A. Natural signs may require immediate self-protective action while official tsunami assessment follows. B. Residents should wait at the beach for confirmation. C. The shaking proves a cyclone is approaching. D. An app can predict the earthquake's exact aftershock sequence.

Answer: A.

Explanation by option:

- **A.** Correct: Natural signs may require immediate self-protective action while official tsunami assessment follows.
- **B.** Wrong: Waiting can consume the near-source action window.
- **C.** Wrong: Cyclone warning uses meteorological, not seismic, evidence.
- **D.** Wrong: Post-event information does not create deterministic prediction.

Trap logic: Near-source natural warning and official warning are complementary, not mutually exclusive.

Q2. Which planning use best represents GIS before a disaster?

A. Using a damage map only after reconstruction is complete. B. Overlaying hazard, population and road layers to site shelters and evacuation routes. C. Issuing an unauthorised public warning from raw satellite imagery. D. Treating a hazard layer as proof of household vulnerability.

Answer: B.

Explanation by option:

- **A.** Wrong: Post-event mapping is a different cycle stage.
- **B.** Correct: Overlaying hazard, population and road layers to site shelters and evacuation routes.
- **C.** Wrong: GIS evidence still requires competent warning authority.
- **D.** Wrong: Risk needs exposure, vulnerability and capacity layers.

Trap logic: Pre-disaster GIS supports decisions; it does not prevent the hazard.

Q3. During a landslide response, which GIS use is most defensible?

A. Declaring the slope permanently safe without field validation. B. Predicting every secondary slide with certainty. C. Combining access, terrain and incident reports to plan search-and-rescue movement. D. Replacing the incident commander with an automated map.

Answer: C.

Explanation by option:

- **A.** Wrong: A map cannot certify long-term safety.
- **B.** Wrong: Uncertainty remains in rapidly changing slope conditions.
- **C.** Correct: Combining access, terrain and incident reports to plan search-and-rescue movement.
- **D.** Wrong: GIS supports, rather than replaces, operational command.

Trap logic: Operational mapping is dynamic and must be checked against field conditions.

Q4. Which post-disaster GIS application is source-grounded?

A. Generating earthquake magnitude before the event. B. Replacing compensation verification with one satellite image. C. Proving that recovery funds were equitably spent. D. Locating heavily damaged areas and guiding reconstruction toward less-vulnerable sites.

Answer: D.

Explanation by option:

- **A.** Wrong: Earthquake magnitude is observed after occurrence.
- **B.** Wrong: Satellite evidence may need ground verification.
- **C.** Wrong: A spatial product does not prove financial or equity outcomes.
- **D.** Correct: Locating heavily damaged areas and guiding reconstruction toward less-vulnerable sites.

Trap logic: Damage identification is not the same as completed rehabilitation.

Q5. Under WMO usage, nowcasting most safely refers to:

A. Detailed present-to-very-short-range weather description extending conventionally to about six hours. B. A deterministic seasonal monsoon prediction. C. A public evacuation order issued by any sensor operator. D. A fixed five-to-thirty-minute window for every service and hazard.

Answer: A.

Explanation by option:

- **A.** Correct: Detailed present-to-very-short-range weather description extending conventionally to about six hours.
- **B.** Wrong: Seasonal prediction is a different forecast horizon.
- **C.** Wrong: Warning authority is separate from forecast production.
- **D.** Wrong: The source's narrow extrapolation example is not the universal definition.

Trap logic: State the WMO range and then qualify the operational product being discussed.

Q6. How should IMD's commonly issued three-hour nowcast be described?

A. As proof that every nowcast worldwide lasts exactly three hours. B. As an operational validity practice within the broader very-short-range nowcasting concept. C. As a tsunami advisory. D. As deterministic earthquake prediction.

Answer: B.

Explanation by option:

- **A.** Wrong: Operational practice does not redefine the global concept universally.
- **B.** Correct: As an operational validity practice within the broader very-short-range nowcasting concept.
- **C.** Wrong: INCOIS assesses tsunami threat.
- **D.** Wrong: Weather nowcasting cannot predict earthquakes.

Trap logic: Do not turn a common operational duration into a universal scientific law.

Q7. The source's three-to-six-hour Doppler/hydromet figure should be used as:

A. A guaranteed evacuation window for every Indian city. B. The detection time for every tsunami. C. A qualified urban-rainfall monitoring and planning lead-time example. D. The prediction horizon for earthquakes.

Answer: C.

Explanation by option:

- **A.** Wrong: Local systems, storm evolution and action needs vary.
- **B.** Wrong: Tsunami assessment uses seismic and sea-level systems.
- **C.** Correct: A qualified urban-rainfall monitoring and planning lead-time example.
- **D.** Wrong: Earthquakes remain non-predictable in this sense.

Trap logic: A source-specific planning figure is not a universal performance guarantee.

Q8. A municipal DSS generates a severe urban-flood map. What must follow before public action?

A. Automatic publication of every raw model run. B. Transfer of CWC's mandate to the software vendor. C. A claim that flooding has already been prevented. D. Competent official interpretation and an authorised warning linked to local action.

Answer: D.

Explanation by option:

- **A.** Wrong: Raw outputs need validation and communication discipline.
- **B.** Wrong: Software does not acquire statutory or institutional mandate.
- **C.** Wrong: A forecast product is an input, not an outcome.
- **D.** Correct: Competent official interpretation and an authorised warning linked to local action.

Trap logic: DSS generation, warning issuance and protective action are separate stages.

Q9. Which message is genuinely impact-based?

A. Intense rainfall may inundate named underpasses and disrupt hospital access; avoid these routes. B. Rainfall may be 80 mm. C. A low-pressure area exists. D. The radar reflectivity is increasing.

Answer: A.

Explanation by option:

- **A.** Correct: Intense rainfall may inundate named underpasses and disrupt hospital access; avoid these routes.
- **B.** Wrong: A parameter alone does not state likely consequence.
- **C.** Wrong: A weather feature alone does not specify local impact.
- **D.** Wrong: A sensor trend is not yet an action message.

Trap logic: Impact-based forecasting translates hazard parameters into consequence and action.

Q10. Why must warning lead time be hazard-specific?

A. All agencies use identical sensors. B. Hazard physics, observation capability and feasible protective actions differ. C. Every household has equal mobility. D. CAP fixes the scientific forecast horizon.

Answer: B.

Explanation by option:

- **A.** Wrong: Sensors and models are hazard-specific.
- **B.** Correct: Hazard physics, observation capability and feasible protective actions differ.
- **C.** Wrong: Social capacity varies and affects usable time.
- **D.** Wrong: CAP standardises message format, not hazard physics.

Trap logic: Usable lead time is scientific and social, not merely a clock value.

Q11. Which statement preserves the earthquake boundary?

A. BhooKamp predicts the next earthquake. B. ITEWS predicts the earthquake that may cause a tsunami. C. Technology can rapidly report earthquake parameters after occurrence but cannot reliably predict exact time, place and magnitude beforehand. D. Crowd reports replace the seismic network.

Answer: C.

Explanation by option:

- **A.** Wrong: BhooKamp is an information application.
- **B.** Wrong: ITEWS detects and assesses after seismic occurrence.
- **C.** Correct: Technology can rapidly report earthquake parameters after occurrence but cannot reliably predict exact time, place and magnitude beforehand.

- **D.** Wrong: Crowd signals need validation and cannot replace seismology.

Trap logic: Do not relabel rapid post-event detection as prediction.

Q12. The ITEWS under-ten-minute capability most directly concerns:

A. Guaranteed delivery to every fisher within ten minutes. B. The local evacuation order in every coastal district. C. The final all-clear after sea-level monitoring. D. Detection and characterisation of potentially tsunamigenic earthquakes across the Indian Ocean.

Answer: D.

Explanation by option:

- **A.** Wrong: Delivery depends on channels and access.
- **B.** Wrong: Protective authority is institutionally distinct.
- **C.** Wrong: All-clear requires continued assessment and observation.
- **D.** Correct: Detection and characterisation of potentially tsunamigenic earthquakes across the Indian Ocean.

Trap logic: Scientific detection is one stage of a longer warning chain.

Q13. Which description of the Cyclone Warning Dissemination System is correct?

A. An IMD satellite-based direct-dissemination arrangement for cyclone-prone coastal areas through cyclone warning centres. B. An NDMA earthquake-prediction network. C. A CWC reservoir-release order. D. An INCOIS tsunami modelling tool.

Answer: A.

Explanation by option:

- **A.** Correct: An IMD satellite-based direct-dissemination arrangement for cyclone-prone coastal areas through cyclone warning centres.
- **B.** Wrong: NDMA does not own cyclone science.
- **C.** Wrong: CWC hydrological functions are distinct.
- **D.** Wrong: INCOIS owns tsunami assessment, not cyclone dissemination.

Trap logic: Attribute the system to IMD and retain its cyclone-specific function.

Q14. What is CWC's principal warning contribution in this topic?

A. Issuing cyclone track forecasts. B. Monitoring river conditions and issuing flood forecasts to administrations and relevant authorities. C. Operating the national earthquake app. D. Authoring all CAP alerts for every hazard.

Answer: B.

Explanation by option:

- **A.** Wrong: IMD handles cyclone forecasts.
- **B.** Correct: Monitoring river conditions and issuing flood forecasts to administrations and relevant authorities.
- **C.** Wrong: NCS/MoES owns earthquake information.
- **D.** Wrong: Competent hazard agencies retain alert authorship.

Trap logic: River-flood forecasting and urban-drainage DSS are related but not identical.

Q15. Which statement correctly limits IMD's role?

A. IMD predicts earthquakes. B. IMD issues all tsunami advisories. C. IMD is the competent meteorological and cyclone-warning agency, not the owner of every hazard warning. D. IMD alone orders every evacuation.

Answer: C.

Explanation by option:

- **A.** Wrong: Earthquake prediction is not available.
- **B.** Wrong: INCOIS assesses tsunami threat.
- **C.** Correct: IMD is the competent meteorological and cyclone-warning agency, not the owner of every hazard warning.
- **D.** Wrong: State and local authorities implement many protective decisions.

Trap logic: Name the agency and the hazard together.

Q16. INCOIS contributes to early warning primarily through:

A. Urban-drain maintenance. B. Cyclone evacuation orders. C. Post-earthquake building-code enforcement. D. Tsunami assessment using seismic, modelling and sea-level information.

Answer: D.

Explanation by option:

- **A.** Wrong: Municipal systems own drainage operations.
- **B.** Wrong: Local authorities order evacuation.
- **C.** Wrong: Construction enforcement is a separate governance function.
- **D.** Correct: Tsunami assessment using seismic, modelling and sea-level information.

Trap logic: INCOIS provides scientific advisories, not every downstream action.

Q17. What is the strongest description of ISRO/NRSC support?

A. Earth-observation and geospatial products that feed risk mapping, response assessment and recovery planning. B. Automatic replacement of district administration. C. A guarantee that satellite images are cloud-free and immediate. D. Sole authority to issue every public warning.

Answer: A.

Explanation by option:

- **A.** Correct: Earth-observation and geospatial products that feed risk mapping, response assessment and recovery planning.
- **B.** Wrong: Geospatial products support institutional decisions.
- **C.** Wrong: Resolution, cloud and processing constraints remain.
- **D.** Wrong: Hazard agencies and governments retain warning authority.

Trap logic: Remote sensing observes; institutions interpret and act.

Q18. What does CAP solve most directly?

A. Scientific ownership of compound hazards. B. Structured message interoperability across multiple dissemination channels. C. Absence of evacuation transport. D. Universal comprehension of any message.

Answer: B.

Explanation by option:

- **A.** Wrong: Mandates remain institutional questions.
- **B.** Correct: Structured message interoperability across multiple dissemination channels.
- **C.** Wrong: Transport is a preparedness capacity.
- **D.** Wrong: Accessibility and trust require additional design.

Trap logic: CAP improves format reuse, not the whole warning chain.

Q19. Which statement about SACHET is safest?

A. It is India's earthquake-prediction centre. B. Its existence proves every at-risk person receives alerts. C. It is NDMA's CAP-based integrated alert portal and dissemination layer. D. It transfers hazard science from IMD and INCOIS to NDMA.

Answer: C.

Explanation by option:

- **A.** Wrong: SACHET is not a prediction centre.
- **B.** Wrong: Platform availability is not receipt.
- **C.** Correct: It is NDMA's CAP-based integrated alert portal and dissemination layer.
- **D.** Wrong: Competent hazard agencies retain scientific functions.

Trap logic: Treat portal existence, operational coverage and human reach as separate claims.

Q20. Cell broadcast differs from individually addressed SMS because it:

A. Predicts local hazards without an agency. B. Works on every handset without status limits. C. Eliminates the need for sirens or local relay. D. Can transmit an area-targeted alert to compatible devices within selected cells.

Answer: D.

Explanation by option:

- **A.** Wrong: It is a delivery channel, not hazard science.
- **B.** Wrong: Compatibility and rollout are separate dated facts.
- **C.** Wrong: Redundant non-digital channels remain necessary.
- **D.** Correct: Can transmit an area-targeted alert to compatible devices within selected cells.

Trap logic: A technical delivery advantage does not prove universal reach.

Q21. Which app-owner pair is correct?

A. Mausam-IMD. B. Mausam-CWC. C. Mausam-INCOIS. D. Mausam-NCS.

Answer: A.

Explanation by option:

- **A.** Correct: Mausam-IMD.
- **B.** Wrong: CWC handles river-flood forecasts.
- **C.** Wrong: INCOIS handles tsunami assessment.
- **D.** Wrong: NCS handles earthquake information.

Trap logic: Do not list applications without institutional attribution.

Q22. Damini is associated with:

A. INCOIS and tsunami advisories. B. IITM Pune and lightning information or warning. C. CWC and river-stage forecasts. D. NDMA and every CAP alert.

Answer: B.

Explanation by option:

- **A.** Wrong: Tsunami science belongs to INCOIS.
- **B.** Correct: IITM Pune and lightning information or warning.
- **C.** Wrong: River forecasting belongs to CWC.
- **D.** Wrong: NDMA integration does not make every app its scientific product.

Trap logic: Lightning applications should not be reassigned to unrelated agencies.

Q23. Which current earthquake-application statement is correct?

A. Both names prove earthquake prediction. B. BhooKamp is an IMD cyclone app. C. BhooKamp is the current NCS/MoES anchor; India Quake is the document-period name in the older source. D. India Quake is an INCOIS tsunami model.

Answer: C.

Explanation by option:

- **A.** Wrong: Both disseminate information rather than deterministic prediction.
- **B.** Wrong: Cyclone services belong to IMD.
- **C.** Correct: BhooKamp is the current NCS/MoES anchor; India Quake is the document-period name in the older source.
- **D.** Wrong: Tsunami modelling belongs to INCOIS.

Trap logic: Preserve the date boundary between document-period and current application names.

Q24. Which EW4All mapping is correct?

A. ITU-risk knowledge, IFRC-forecasting, WMO-telecom, UNDRR-response. B. One pillar led solely by NDMA. C. A completed 2027 universal-coverage certificate. D. UNDRR-risk knowledge, WMO-detection, ITU-dissemination, IFRC-preparedness and response.

Answer: D.

Explanation by option:

- **A.** Wrong: The lead organisations are assigned to different pillars.
- **B.** Wrong: EW4All is global and multi-agency.
- **C.** Wrong: The 2027 date is a target, not a completion statement.
- **D.** Correct: UNDRR-risk knowledge, WMO-detection, ITU-dissemination, IFRC-preparedness and response.

Trap logic: Use the four pillars as an assessment framework.

Q25. How should Cyclone Phailin be used as evidence?

A. As accurate forecasting and tracking combined with preparedness and evacuation. B. As proof of deterministic cyclone prediction. C. As evidence that technology alone prevented all loss. D. As proof that all hazards provide days of warning.

Answer: A.

Explanation by option:

- **A.** Correct: As accurate forecasting and tracking combined with preparedness and evacuation.
- **B.** Wrong: Forecasts remain probabilistic.
- **C.** Wrong: Administration and community action mattered.
- **D.** Wrong: Hazard lead times differ.

Trap logic: Credit the chain, not a single technological input.

Q26. Why are detection and warning issuance separable?

A. Every sensor operator is a public authority. B. A sensor signal requires validation, interpretation and competent authorisation before public warning. C. Detection automatically ensures receipt. D. Forecast models also order evacuation.

Answer: B.

Explanation by option:

- **A.** Wrong: Technical operators may lack warning mandate.
- **B.** Correct: A sensor signal requires validation, interpretation and competent authorisation before public warning.
- **C.** Wrong: Delivery and receipt are downstream stages.

- **D.** Wrong: Protective authority is institutionally distinct.

Trap logic: A strong detection layer can coexist with an issuance bottleneck.

Q27. Which indicator best tests equitable last-mile reach?

A. Total number of app downloads. B. National area shown on a portal. C. Receipt, comprehension and action rates disaggregated by language, disability, gender and location. D. Number of alerts authored.

Answer: C.

Explanation by option:

- **A.** Wrong: Downloads do not show timely receipt or action.
- **B.** Wrong: Geographic availability does not show social reach.
- **C.** Correct: Receipt, comprehension and action rates disaggregated by language, disability, gender and location.
- **D.** Wrong: Authorship is an administrative input.

Trap logic: Measure the people most likely to be excluded.

Q28. What is the purpose of warning-system redundancy?

A. To issue contradictory messages from many agencies. B. To remove all maintenance needs. C. To make local volunteers unnecessary. D. To preserve essential warning functions when a sensor, network, power source or channel fails.

Answer: D.

Explanation by option:

- **A.** Wrong: Consistency and authority remain necessary.
- **B.** Wrong: More components still require maintenance.
- **C.** Wrong: Human relay is part of redundancy.
- **D.** Correct: To preserve essential warning functions when a sensor, network, power source or channel fails.

Trap logic: Redundancy should avoid both single points of failure and message conflict.

Q29. Which is an appropriate use of drones in disaster management?

A. Rapid local imagery for access or damage assessment under airspace, safety and validation controls. B. Unreviewed public warning based on one image. C. Permanent surveillance unrelated to the emergency. D. Replacement of all ground assessment.

Answer: A.

Explanation by option:

- **A.** Correct: Rapid local imagery for access or damage assessment under airspace, safety and validation controls.
- **B.** Wrong: One image needs context and authority.
- **C.** Wrong: Purpose limitation is a privacy safeguard.
- **D.** Wrong: Ground validation remains necessary.

Trap logic: Operational speed does not remove safety, privacy or evidence limits.

Q30. What is a necessary control for AI-assisted prioritisation?

A. Treating model output as final truth. B. Human oversight, provenance, bias testing and uncertainty disclosure. C. Training only on easily connected populations. D. Keeping no record of data source or version.

Answer: B.

Explanation by option:

- **A.** Wrong: Models can be wrong or incomplete.
- **B.** Correct: Human oversight, provenance, bias testing and uncertainty disclosure.
- **C.** Wrong: Digital data can systematically exclude groups.

- **D.** Wrong: Auditability requires provenance.

Trap logic: AI should assist accountable decisions, not obscure them.

Q31. How should crowdsourced disaster reports be handled?

A. Publish every report as verified warning. B. Discard all community information. C. Use them as rapid local signals after checking source, time, location and duplication. D. Use reports to replace competent hazard agencies.

Answer: C.

Explanation by option:

- **A.** Wrong: False or stale reports can mislead.
- **B.** Wrong: Community data can add valuable local detail.
- **C.** Correct: Use them as rapid local signals after checking source, time, location and duplication.
- **D.** Wrong: Scientific and public authority remain necessary.

Trap logic: Crowdsourcing adds speed and local knowledge only with verification.

Q32. Which statement expresses the technology-outcome firewall?

A. Technology deployment automatically proves resilience. B. Every forecast causes evacuation. C. A dashboard eliminates vulnerability. D. A platform or alert proves an input; timely action and avoided loss require separate evidence.

Answer: D.

Explanation by option:

- **A.** Wrong: Inputs and outcomes must be evaluated separately.
- **B.** Wrong: Protective decisions involve authority and capacity.
- **C.** Wrong: Social and physical vulnerability persist.
- **D.** Correct: A platform or alert proves an input; timely action and avoided loss require separate evidence.

Trap logic: Never convert a tool, allocation or alert count directly into a success claim.

PYQS AND ANSWER PRACTICE

Verified PYQ routing rule

No verified question below directly owns the whole multi-hazard early-warning architecture. Each card reproduces the verified demand and solves only the early-warning or disaster-technology contribution; the primary topic remains identified.

2026 GS-III Q7 - support application

Verified question: "Discuss how the contradiction between 'rapid infrastructure development' and 'disaster-risk reduction' in ecologically-sensitive areas of India can be managed, with suitable examples." **10 marks · 150 words.**

Primary owner: Landslides, Avalanches and GLOF Risk; critical-infrastructure resilience also supports it.

Topic 04 application solution: Risk-sensitive infrastructure needs an observation-to-action loop across its lifecycle. Before approval, GIS and remote sensing should combine hazard, slope, drainage, exposure and cumulative-development layers to test alignment and alternatives. During construction and operation, rainfall, river, slope, glacial-lake or seismic observations should feed threshold-based monitoring and competent warnings. Alerts must trigger pre-agreed actions-traffic closure, reservoir or work-site protocols, evacuation and inspection-rather than remain dashboard outputs. Redundant sensors, backup power, independent audits and community reporting reduce single-point failure. However, monitoring cannot legitimise construction in an unacceptable-risk location; carrying capacity, ecosystem buffers, enforceable siting and design remain primary. Thus technology manages residual risk only when embedded in precautionary planning and accountable lifecycle governance.

2026 GS-III Q18 - support application

Verified question: "'Community participation is the cornerstone of effective disaster management.' Analyse this statement with suitable examples from India. Also discuss the challenges to community participation and measures to strengthen it." **15 marks · 250 words.**

Primary owner: Community-Based DRR and Inclusive Protection.

Topic 04 application solution: Community participation closes the warning system's last mile. Residents contribute local hazard history, identify people needing assistance, validate routes and shelters, relay official alerts and report whether messages were understood and acted upon. Odisha's cyclone preparedness illustrates how forecast skill gains protective value through evacuation practice and local mobilisation. Participation also exposes digital exclusion: app or cell-broadcast availability does not ensure access for persons with disabilities, migrants, non-literate groups or households without reliable devices. Challenges include token consultation, volunteer fatigue, mistrust after false alarms, unequal representation and unclear authority. Strengthening measures include ward- and village-level risk mapping, multilingual and accessible messages, trained and insured volunteers, recurring drills, feedback indicators disaggregated by group, and formal integration of community relays with district control rooms. Community knowledge should complement, not replace, competent scientific agencies and accountable public authority.

2024 GS-III Q17 - support application

Verified question: "What is disaster resilience? How is it determined? Describe various elements of a resilience framework. Also mention the global targets of Sendai Framework for Disaster Risk Reduction (2015-2030)." **15 marks · 250 words.**

Primary owner: Concepts, Risk, Resilience and Sendai.

Topic 04 application solution: Early warning contributes to resilience through six linked elements: updated risk knowledge; hazard-specific observation and forecasting; authoritative warning; interoperable and redundant dissemination; inclusive preparedness and action; and feedback that corrects maps, thresholds and communication. Resilience is not determined by sensor density alone. It depends on whether alerts reach exposed groups, are understood and trusted, and trigger feasible action while critical services remain functional. The Sendai targets most directly connected to this topic are reducing mortality, affected people, economic loss and critical-infrastructure disruption, while increasing access to multi-hazard early-warning systems and disaster-risk information. EW4All's four pillars-risk knowledge, detection, dissemination, and preparedness-provide a current implementation lens, but its 2027 target is not proof of universal coverage.

2024 GS-III Q18 - support application

Verified question: "Flooding in urban areas is an emerging climate-induced disaster. Discuss the causes of this disaster. Mention the features of two major floods in the last two decades in India. Describe the policies and frameworks in India that aim at tackling such floods." **15 marks · 250 words.**

Primary owner: Riverine Floods and Urban Flood Resilience.

Topic 04 application solution: The early-warning contribution combines IMD rainfall observation and nowcasting, CWC river information where relevant, local gauges, Doppler radar, drainage and terrain data, and a municipal decision-support system. Impact-based messages should identify likely inundated roads, service disruption and required action rather than communicate rainfall alone. The source's three-to-six-hour hydrometeorological figure is a qualified planning example, not a universal guarantee. Government-authorised warnings should move through CAP/SACHET, broadcast, sirens and local relays, with traffic control, pumping, evacuation and shelter triggers agreed beforehand. GIS should continuously update low-lying settlements, hospitals, underpasses and drainage obstructions. This technology layer cannot substitute for wetland protection, storm-water design, land-use control or maintenance-the principal causes and policy framework remain Topic 08's responsibility.

2020 GS-III Q15 - technology-governance application

Verified question: "The COVID-19 pandemic has caused unprecedented devastation worldwide. However, technological advancements are being availed readily to win over the crisis. Give an account of how technology was sought to aid management of the pandemic." **15 marks · 250 words.**

Primary owner: Epidemics and Public-Health Emergencies.

Topic 04 application solution: The transferable warning-system lesson is an information-to-action chain: surveillance data require quality control; models require expert interpretation; authorised messages require consistent multi-channel dissemination; and recipients need feasible action. Interoperable data and dashboards can improve situational awareness, while applications and geo-targeting can speed communication. Yet technology also creates false-alarm, privacy, exclusion and accountability risks. Necessity, data minimisation, access control, provenance, human oversight and non-digital alternatives are therefore essential. The pandemic-specific technologies and outcomes belong to the public-health topic; this chapter contributes the governance test that a platform or message is an input, not proof of equitable uptake or reduced loss.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why a multi-hazard early-warning system must be people-centred and end-to-end. **150 words.**

Model answer (137 words): A multi-hazard early-warning system is not a sensor network but a chain that converts risk knowledge into protective action. First, hazard, exposure, vulnerability and capacity data identify who is at risk and what action is possible. Second, hazard-specific agencies observe and forecast: IMD for weather and cyclones, CWC for river floods and INCOIS for tsunami threat. Third, a competent authority issues a clear warning with location, timing, uncertainty and action. Fourth, redundant channels-CAP/SACHET, broadcast, sirens and local relays-carry one consistent message. Fifth, preparedness supplies routes, shelters, transport, drills and assistance. Finally, feedback tests receipt, comprehension, action and avoided loss. Multi-hazard integration can share data and dissemination standards, but it cannot erase different hazard physics, mandates or lead times. The system is people-centred because success is measured by timely inclusive action, not alerts issued.

ORIGINAL MAINS 2 - 10 MARKS

Question: Describe the role of GIS and remote sensing across the disaster-management cycle. **150 words.**

Model answer (134 words): Remote sensing provides repeated, synoptic observations; GIS integrates those observations with terrain, settlements, roads, hospitals, shelters and demographic data. Before a disaster, GIS supports vulnerability mapping, route planning, no-construction decisions and shelter siting-illustrated by cyclone-shelter planning in Odisha. During response, current imagery and field reports help identify blocked access, likely spread and search-and-rescue priorities, as in the Sikkim landslide example. After impact, damage mapping can prioritise relief and guide reconstruction toward safer sites, as used in the Gujarat earthquake region. These tools improve spatial decisions but do not themselves issue authoritative warnings or prove that rescue and recovery occurred. Cloud cover, resolution, processing delay, outdated layers and weak ground validation can distort decisions. Therefore remote sensing should observe, GIS should integrate, competent institutions should decide and field teams should validate and act.

ORIGINAL MAINS 3 - 15 MARKS

Question: Compare warning capability and lead time across major hazards in India. **250 words.**

Model answer (211 words): Warning capability differs because hazards have different physical processes and observing systems. Earthquakes cannot be reliably predicted in exact time, place and magnitude; seismic networks and NCS applications rapidly disseminate parameters after occurrence. A potentially tsunamigenic earthquake can, however, trigger assessment: INCOIS's ITEWC combines seismic information, models, Bottom Pressure Recorders and tide gauges, with the source recording under-ten-minute earthquake detection-not guaranteed evacuation. Severe weather permits nowcasting: WMO usage extends through roughly six hours, while IMD commonly issues three-hour operational nowcasts. The source's five-to-thirty-minute extrapolation and three-to-six-hour Doppler/hydrometeorological urban-rainfall figure should be treated as qualified service examples. Cyclones can provide longer tracking windows through IMD satellites, radar and models; Phailin demonstrates forecast-plus-preparedness rather than deterministic prediction. River floods depend on catchment, gauges, reservoir

operation and CWC forecasts, whereas flash and urban floods may sharply compress action time.

The policy implication is to match action to usable lead time. Near-zero contexts require resilient construction, automatic shutdowns and immediate self-protection; minute-to-hour warnings require pre-authorised routes and closures; longer cyclone windows permit mass evacuation and pre-positioning. One universal lead-time claim, sensor or agency is therefore analytically wrong.

ORIGINAL MAINS 4 - 15 MARKS

Question: Evaluate CAP, SACHET and cell broadcast as solutions to India's last-mile warning challenge. **250 words.**

Model answer (210 words): CAP, SACHET and cell broadcast solve related but different dissemination problems. Common Alerting Protocol allows a competent hazard agency to author one structured message that can be rendered across SMS, cell broadcast, applications, web, radio, television and sirens. NDMA's SACHET provides an integrated national alert portal and dissemination layer. Cell broadcast can transmit an area-targeted message to compatible devices in selected cells without individually addressing subscribers, reducing SMS queuing constraints.

Their value is speed, consistency, geo-targeting and channel interoperability. However, CAP does not decide which agency owns a compound hazard or whether an evacuation should be ordered. SACHET's existence does not prove universal operational reach. Cell broadcast still depends on network configuration, compatible devices, accessible language, power and user attention. None solves mistrust, disability access, absent transport or shelter, or the exclusion of people without digital connectivity.

India therefore needs one authoritative alert across redundant digital, broadcast and human pathways; multilingual audio, text and visual formats; backup power; local volunteers; and performance metrics for delivery, comprehension and action by social group. These technologies strengthen the last mile, but only preparedness and accountable local authority convert delivery into protection.

ORIGINAL MAINS 5 - 20 MARKS

Question: "India's early-warning capability is stronger at detection than at the last mile." Critically examine. **250 words.**

Model answer (239 words): India possesses substantial hazard-detection capability: IMD integrates satellites, Doppler radar, observations and models for weather and cyclones; CWC monitors river conditions; INCOIS combines seismic, model, Bottom Pressure Recorder and tide-gauge information for tsunami threat; ISRO/NRSC provide earth-observation and geospatial products. Phailin illustrates how strong forecasting can support large-scale protective action.

Yet detection is only the first half of protection. A signal must be validated, converted into an authoritative impact message, disseminated redundantly, understood and trusted, and linked to feasible action. Last-mile failure can arise from delayed authorisation, power or network loss, inconsistent messages, language and disability barriers, digital exclusion, absent routes, transport or shelters, and weak drills. Portal availability or app downloads therefore cannot establish resilience.

The contrast should not be exaggerated. CAP-based SACHET, cell-broadcast capability, public applications and community relays strengthen dissemination, while states with mature cyclone preparedness demonstrate institutional integration. Detection itself also has maintenance, coverage and uncertainty limits, illustrated by vulnerable ocean-observation platforms.

Reform should map performance against EW4All's four pillars: update risk data; maintain diverse sensors and backup systems; clarify thresholds and agency roles; use one accessible message across digital and human channels; finance local preparedness; and measure delivery, comprehension, action and false alarms by group. Thus India's central task is not technology acquisition alone but converting scientific capability into inclusive, rehearsed and measurable early action.

ORIGINAL MAINS 6 - 20 MARKS

Question: Discuss the opportunities and governance risks of drones, artificial intelligence and crowdsourcing in disaster warning and response. **250 words.**

Model answer (232 words): Emerging technologies can compress decision time. Drones provide local imagery when roads are blocked; AI can detect patterns, classify damage or prioritise areas; and crowdsourcing can reveal rapidly changing needs and hazards. Combined with satellites, sensors and GIS, they improve situational awareness, search planning and resource allocation.

Their limitations are equally important. Drone operations face airspace, safety, endurance and privacy constraints. AI inherits missing or biased data, can produce opaque false confidence and may under-represent poorly connected communities. Crowdsourced reports can be false, duplicated, stale or wrongly located. Geo-targeted and device data can become unnecessary surveillance, while digital-only systems exclude people without devices, literacy, connectivity or accessible interfaces.

Governance should therefore require purpose limitation, necessity, data minimisation, access control, retention limits and cybersecurity. Every input needs provenance-source, location, timestamp and version-plus human validation and uncertainty disclosure. Agencies should audit model performance across regions and social groups, establish liability and correction procedures, and preserve non-digital channels and community knowledge. Drones and AI should support, not replace, competent scientific agencies, incident command or field verification.

The correct test is an outcome firewall: a model, image, report or dashboard proves a technical input; timely, equitable action and avoided loss need separate evidence. Responsible innovation combines speed with accountability, privacy and inclusion.